

SCOPE OF WORK (SOW) — CYCLISTIC BIKE-SHARE CASE STUDY

1. Project Title

Cyclistic Bike-Share Analysis: Understanding Usage Differences Between Annual Members and Casual Riders.

2. Scope of Work Summary:

In this phase, I defined the key business question: *How do annual members and casual riders use Cyclistic bikes differently?* I clarified the project's objectives, identified stakeholders, outlined success criteria, and established project scope and deliverables.

These foundations ensure that subsequent analysis directly supports Cyclistic's marketing strategy to convert casual riders into annual members.

3. Background / Business Context

Cyclistic is a bike-share program based in Chicago with over 5,800 bicycles and 600 docking stations. The company's marketing team aims to increase profitability by converting casual riders into annual members, as annual members have proven to be more profitable and consistent users.

As a junior data analyst on the Cyclistic marketing analytics team, I've been assigned to analyse historical bike trip data to understand how annual members and casual riders use Cyclistic bikes differently.

The insights from this analysis will inform future marketing strategies and campaigns targeted at converting casual riders into annual members.

4. Problem Statement

Cyclistic's marketing director believes the company's future success depends on maximizing the number of annual memberships. However, the marketing team lacks data-backed insights into how casual riders and annual members behave differently.

5. Objectives / Key Questions (Ask Phase)

Primary Objective:

To identify behavioural and usage differences between annual members and casual riders based on historical trip data.

To develop strategies to convert more casual riders into annual members.

Thus, this project seeks to answer the following questions:

-  How do annual members and casual riders use Cyclistic bikes differently?
-  Why would casual riders buy Cyclistic annual memberships?
-  How can Cyclistic use digital media to influence casual riders to become members?

6. Scope Definition

In Scope

- ✚ Use Cyclistic historical trip data (12 months of data).
- ✚ Perform data cleaning, transformation, and descriptive analysis.
- ✚ Compare user types (casual vs. member) across trip duration, frequency, time, and station usage.
- ✚ Create visualizations summarizing findings.
- ✚ Provide actionable recommendations for marketing strategy.

Out of Scope

- ✚ Predictive modelling or forecasting.
- ✚ Customer segmentation using external data.
- ✚ Real-time dashboard automation or live data pipelines.

7. Stakeholders

- ✚ **Primary Stakeholder (Fictional):** Lily Moreno, Director of Marketing
- ✚ **Data Analyst:** *Emmanuel Olugbodi (myself)*
- ✚ Cyclistic marketing analytics team
- ✚ **Audience:** Cyclistic Executive Team (fictional, for presentation purpose)

8. Success Criteria

The project will be considered successful when:

- ✚ The analysis clearly distinguishes behavioural differences between rider types.
- ✚ Visualizations effectively communicate usage trends.
- ✚ Insights directly support marketing strategy recommendations.
- ✚ The final report and dashboard are presented in a professional format.

9. Deliverables

Deliverable	Format	Tool
Scope of Work	PDF	Microsoft Word
Visualization Dashboard	.xlsx	Excel
Documentation	PDF	Microsoft Word
Final Presentation Slides	.pptx	Canva

10. Timeline

Phase	Tasks	Duration
Ask	Define problem, objectives, and success metrics	3 days
Prepare	Acquire and inspect data	2 days
Process	Clean, merge, and transform data	3 days
Analyse	Perform EDA, identify patterns	3 days
Share	Visualize insights, summarize findings	3 days
Act	Develop recommendations, finalize report	3 days
Total Estimated Duration:		~17 days

11. Detailed Phase-by-Phase Plan

A. ASK PHASE

- ✚ Review case study brief and define the analytical problem.
- ✚ Identify key stakeholders and project goals.
- ✚ Establish success metrics and deliverables.

B. PREPARE PHASE

- ✚ Download Cyclistic bike trip data from official data source.
- ✚ Understand data structure and key variables
- ✚ Check data completeness and reliability.

C. PROCESS PHASE

- ✚ Clean and transform raw data (handle nulls, date formatting, remove duplicates).
- ✚ Create calculated fields
- ✚ Merge quarterly data into one dataset.

Output: Cleaned and ready-to-analyse dataset.

D. ANALYZE PHASE

- ✚ Conduct descriptive analysis and summary statistics.
- ✚ Identify usage patterns by rider type (trip duration, start time, station).
- ✚ Compare ride frequency, preferred days, and peak hours.

Output: Key Findings with Interpretation

E. SHARE PHASE

- ✚ Create data visualizations and summary dashboard.
- ✚ Prepare visual report and presentation.
- ✚ Ensure clarity, storytelling, and stakeholder relevance.

Output: Dashboard + Summary Insight.

F. ACT PHASE

- ✚ Provide actionable recommendations for converting casual riders into members.
- ✚ Suggest marketing strategies based on data insights.
- ✚ Reflect on project outcomes and potential next steps.

Output: Final report and presentation slide deck.

12. Tools & Technologies

- ✚ **Data Handling & Analysis:** Excel
- ✚ **Visualization:** Excel
- ✚ **Documentation:** Microsoft Word

13. Risks & Mitigation

Risk	Likelihood	Mitigation
Missing or inconsistent data	Medium	Apply data cleaning and outlier checks
Tool limitations (free version)	Low	Use open-source or trial versions
Time management	Medium	Allocate daily milestones for each phase

14. Expected Outcomes

- ✚ Clear comparison of member vs. casual rider behaviour.
- ✚ Data-backed recommendations for marketing strategy.
- ✚ Portfolio-ready report and dashboard demonstrating analytical and visualization skills.

15. References / Data Sources

- ✚ **Source:** [Divvy/Cyclistic public dataset](#)
- ✚ **Case Study Framework:** Google Data Analytics Capstone
- ✚ **Tools:** Microsoft Excel.